Terrestrial Radius, Frenet-Serret Curvature, and Empirical Measurement Sensitivity: A Comment on Flat Earth Theory

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Abstract

We will assess how to relate declination angle measurements to both terrestrial radius and Frenet-Serret curvature of a meridian line. Then we assess why certain empirical methods are numerically instable and alternative methodologies for robust measurement.

1 Empirical Measurement of Terrestrial Radius

We will not assume oblate spheroidal geometry of the Earth. Additionally, we will analyze the curvature along a meridian. Geometrically, this is the intersection of a plane along a sphere.

The arclength s can be obtained [1]:

$$s = \theta r \tag{1}$$

Where θ is the turning angle in radians, and r is the radius.

Assuming an observer at height h above a tangent line to the point O lying on a meridian line of the earth, we obtain:

$$r = \frac{\Delta s}{\frac{\pi}{2} - \alpha} \tag{2}$$

Where α is the declination angle from the observer to the vanishing point of the meridian on their horizon and Δs is the distance from the tangent point O of the observer to the vanishing point H.

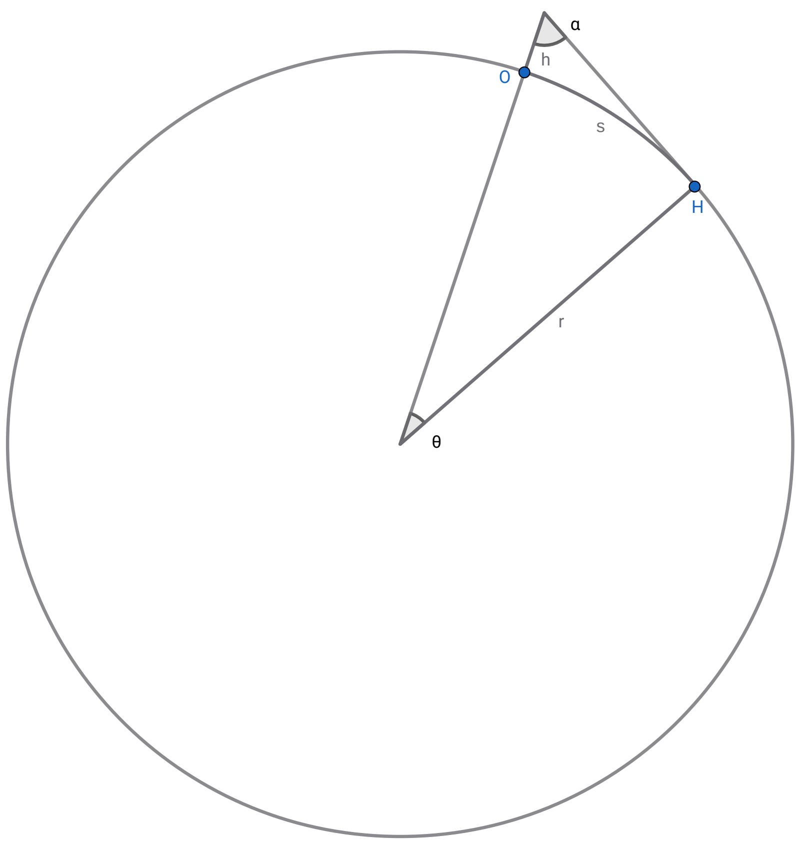

Figure 1: Meridian line with rotating angle θ, declination angle α, observer height h, radius r, and arclength s.

If we define Frenet-Serret curvature k in terms of our turning angle θ, then we have:

$$k = \frac{d\theta}{ds} \tag{3}$$

For empirical measurements of turning angle $\Delta\theta$ and arclength Δs, we have:

$$k \approx \frac{\Delta\theta}{\Delta s} \tag{4}$$

Thus, if the turning angle is below measurement sensitivity then one could infer a curvature $k = 0$, thus suggesting a flat Earth.

2 A Recommendation for Future Measurement Procedure: Al-Biruni

At $h \approx 2$ m, our declination angle is 89.95 deg. The variance of the measurement induced by the local topology of the earth likely produces numerical instability.

The author of this paper suggests that future empirical measurements follow al-Biruni's methodology [2]. The observer should measure declination angle using an astrolabe or similar apparatus, the measurement should be made on top of a mountain, and the landscape should be flat (low topological variance. This will increase measurement robustness.

References

[1] Archimedes and J.L Heiberg. *Archimedis Opera omnia*. 1910.

[2] Muḥammad ibn Aḥmad Bīrūnī. *The Determination of the Coordinates of Positions for the Correction of Distances Between Cities*. The American University of Beirut, 1967, pp. 184–185.